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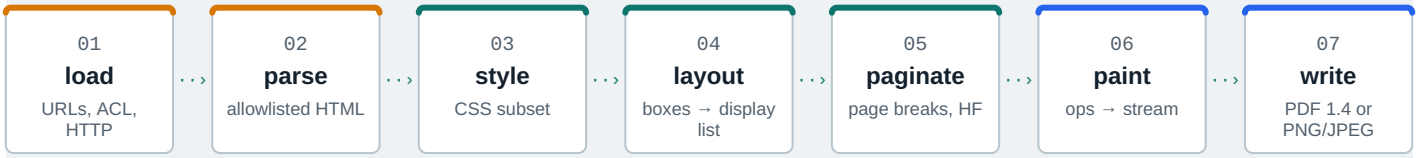
# gowkhtmltopdf — Architecture

A work-alike for *wkhtmltopdf* built for controlled reports: invoices, statements, tables, multi-page documents with headers/footers, TOCs and PDF outlines. Everything from load to write runs inside the Go binary.

| DOCS | DOMAINS | SCALE | COMMIT |
|---|---|---|---|
| 11 deep-dives in documentation/architecture/ | 10 sections on this page | ~71 600 lines of Go | ef526f9 2026-08-11 |

INPUTS ····· SHARED ENGINE ····· OUTPUTS

## Conversion pipeline one document, seven stages




## Reading order jump to any domain



## 01 Entrypoints & CLI

Three entry points: two CLIs and the library; every flag funnels through dotted settings into the convert.Request seam.

The outermost shell turns raw process arguments into a settings-shaped `cli.Command` and hands it to the application adapters that build engine requests. One parser serves both binaries through a `Mode` bitmask, and the parser is the only place that understands the wkhtmltopdf-compatible multi-object grammar: `page`, `cover` and `toc` objects plus free positionals.

ONE PARSER, TWO BINARIES — ARGV IN, REQUEST OUT

```
$ gowkhtmltopdf --page-size A4 cover cover.html toc page ch1.html out.pdf
write PDF: 3 pages · outline depth 2 · exit 0
```

EXIT-CODE CONTRACT (WKHTMLTOPDF COMPATIBLE)

| CODE | MEANING | SOURCE |
|---|---|---|
| `0` | ok | success |
| `1` | error | parse or engine failure |
| `2` | HTTP 404 | settings.HttpStatusError |
| `3` | HTTP 401 | settings.HttpStatusError |

Flags land in settings through dotted names: `--page-size` writes *`size.pagesize`*; free positionals become *`Command.Objects`* and the last one becomes *`Output`*. Errors surface as `gowkhtmltopdf: unknown option --dpi` on stderr with exit code 1, while HTTP failures report **`2`** or **`3`**. Long option names may split at word boundaries when printed.

`argv` → `cli.Parse` → `Command` → `app adapter` → `convert.Request`

| | |
|---|---|
| **`cmd/gowkhtmltopdf`** | PDF binary main — parse (ModePDF), doc-flag dispatch, signals, exit code; 83 thin lines |
| **`cmd/gowkhtmltoimage`** | Image binary main — parse (ModeImage), no TOC-XSL dump, exit code; 53 thin lines |
| **`internal/cli`** | Parser core: Command, Parse, Mode bitmask, ~71 registered flags, ExitCode mapping |
| **`internal/app`** | Adapters BuildPDFRequest / RunImage turn Command into engine requests |

- `cli.Command` — parse result: Global/Image settings, Objects, Output / OutputWriter
- `cli.Parse(argv, Mode)` — token loop, object grammar, pending remapping
- `cli.ExitCode(err)` — 0/1/2/3 via HttpErrorCode
- `settings.Set("size.pagesize", ...)` — the dotted settings home appliers write
- `newFreshObject()` — TOC objects never consume pending — no ghost page

> Security: local-file ACL is opt-in (`--enable-local-file-access` dual-writes *`load.blocklocalfileaccess`*); Policy A rejects inert wkhtmltopdf stubs (`--dpi`, `--javascript-delay`) as unknown rather than no-ops; validation precedes output truncation; the CLI spawns no subprocess and cancels cleanly on SIGINT/SIGTERM via a context.

Position: argv → settings → Request (left edge of the pipeline) · see documentation/architecture/01-entrypoints-cli.md

## 02 Public library API

ROOT PACKAGE

Converter / ImageConverter public API: typed and dotted dialects, deep-copy ownership, in-memory output.

The root package `gowkhtmltopdf` is the idiomatic, stdlib-only Go surface for the engine. It configures a conversion with either wkhtmltopdf-compatible dotted strings or compile-time-discoverable typed builders, and it drives the pipeline: `Converter.Convert(ctx)` reduces to an internal `convert.Request` handed to `convert.Run` (PDF) or `imageout.RunRequest` (image), with all output captured in memory. This package is also the module boundary that enforces the product rule — `pure Go, no cgo, stdlib + internal only` — so any new external dependency would have to cross its import list first.

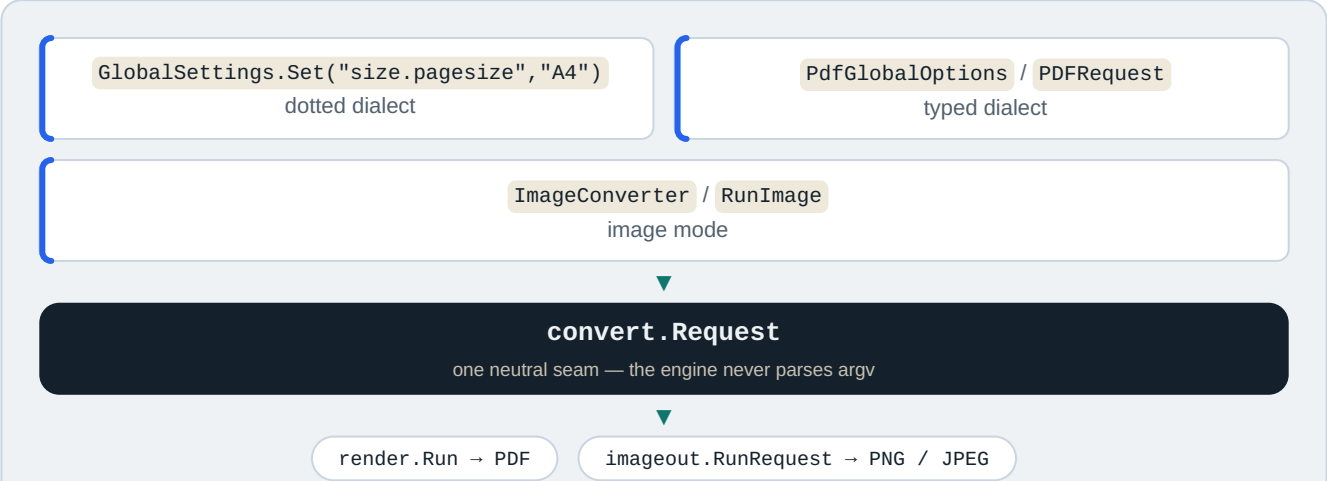



fig. 2 — two dialects, one seam. *api.go* (960 lines) never imports `internal/cli`; both frontends meet at the shared request union (2026-08-11, commit ef526f9).

| | |
|---|---|
| **api.go** | all exported types: settings wrappers, converters, typed requests, hooks, sentinels |
| **api_test.go** | round trips, deep-copy mutation isolation, cancellation, callbacks, PNG/JPEG decode |
| **doc.go** | quick start, dotted settings, ACL pair, in-memory source kind, thread-safety contract |
| **internal/convert, settings, imageout, line, errs** | the five internal dependencies the root package adapts |

- `Converter` + `ImageConverter` — lifecycle: `New`, `AddObject`, `Convert(ctx)`, `Output()`
- `GlobalSettings` / `ObjectSettings` — dotted `Set` / `Get` via the reflect key tables
- `PDFRequest` / `ImageRequest` + `RunPDF` / `RunImage` — typed one-shot; `Now func() time.Time` gives byte-stable output
- `ConvertHTML` — one-shot helper: clone global, `SetBody(html, base)`, convert, output
- `Version()` — banner `0.12.7-dev (gowkhtmltopdf pure-go)`, tracking LibraryVersion 0.12.7-dev
- Sentinels — `ErrNoPageObjectsAdded`, `ErrEmptyHTML`, `ErrNilContext`, matchable with `errors.Is`
- `convertHooks` + `lineLog` — engine log lines classified by `line.SeverityOf` into `OnInfo` / `OnWarn` / `OnError`

| Surface | Contract |
|---|---|
| `Converter` | PDF, multi-object, `OnPhase` / `OnProgress` (final 100); no `Now` hook |
| `ImageConverter` | PNG/JPEG, single most-recent page, lazy zero-value init, no phase callbacks |
| `RunPDF` / `RunImage` | typed path — `Now` injectable for reproducible metadata |

> **Security posture:** local-file ACL is default-deny and needs the same two-step opt-in as the CLI (`enablelocalfileaccess=true` + `load.blocklocalfileaccess=false`); `SetBody` / `ConvertHTML` bypass URL guessing so in-memory HTML cannot be coerced into SSRF; context cancellation threads through every load. Limits: no `HttpErrorCode()` on `Converter` (CLI-only today), no TOC as a first-class object, one image page only. Every boundary copies: `SetBody`, `AddObject`, `Output`.

Same seam as the CLIs: `api.go → convertHooks → Request → render.Run` — deep-dive: `documentation/architecture/02-library-api.md`

## 03 Settings system & errors

wkhtmltopdf-style settings model with reflection key tables; settings and errs are leaf packages.

`internal/settings` is the single source of truth for every user-facing knob — page geometry, margins, headers/footers, TOC behaviour, load policy, image output — and the only package that speaks the wkhtmltopdf dotted-name vocabulary (`margin.top`, `load.jsdelay`, `web.background`). It sits upstream of everything: argv and library calls funnel through `Set("margin.top", "25mm")`, and every engine package reads the typed structs it default-initialises. Its companion leaf `internal/errs` owns the shared sentinel errors across the library boundary.

### DOTTED-KEY PATH — ONE KNOB, TYPED ALL THE WAY DOWN

`--margin-top 25mm` → `Set("margin.top", "25mm")` → `Margin.Top = 25 mm` → `10 mm → 28.346 pt`

Booleans coerce liberally (`"" / true / 1 / yes / on`); unknown keys fail loudly — a typo like `margin.topx` errors instead of degrading.

### POLICY A — SETTINGS HONESTY

Only options with an engine consumer get typed fields. Inert wkhtml keys (`dpi`, `javascript`, `js-delay`, `produce-forms`, `default-encoding`, .) sink into an `Ignored` map: *accepted* so existing invocations do not hard-fail, *inert* so they are never silently promoted to typed stubs. Origin: the 2026-08-06 ponytail review, phase 1.

accepted-inert wkhtml keys, round-trippable via Get

ISO/ANSI page sizes — A4 = 595.28 × 841.89 pt

| | |
|---|---|
| `internal/settings/settings.go` | Typed model: `PdfGlobal`, `PdfObject`, `ImageGlobal` plus `Web`, `LoadGlobal`, `LoadPage`, `HeaderFooter`, `TableOfContent` — wkhtmltopdf-compatible defaults (A4 portrait, 10 mm margins, depth-4 outline). |
| `internal/settings/reflect.go` | Descriptor engine: dotted `Set`/`Get`, type-coercing setters, ignored-key tables — built once at init, zero per-call reflection cost. |
| `internal/settings/unitreal.go · pagesize.go · httperror.go` | `UnitReal` (`10mm`, `1.5in`, `12pt`, `96px`, `100%`), the 23-entry page-size table, and `HttpStatusError` (404→2, 401→3, else 1). |
| `internal/errs/errs.go` | Shared sentinels: `ErrNilContext`, `ErrNilLoader`, `ErrNilCommand`, `ErrNilRequest` — re-exported by `api.go` and `internal/app`. |

- `PdfGlobal / PdfObject / ImageGlobal` — the three settings payloads carried by `cli.Command` and `convert.Request`.
- `Set / Get` — dotted-key surface; accepted inert keys round-trip, unknown keys error.
- `UnitReal` — unit-suffix scalar; `Points()` / `Mm()` conversion.
- `ParsePageSize` — case-insensitive ISO/ANSI names to points; empty string → A4.
- `HttpErrorCode` — 404→2, 401→3, else 1; consumed by `cli.ExitCode` via an interface check.

### SECURITY & LIMITS

Local file access is denied by default (`BlockLocalFileAccess: true`); `--allow` prefixes are the only ACL widening; proxies must be absolute `http(s)` URLs. Inert-key acceptance is a compat surface, not an attack surface — the engine never executes scripts or creates forms.

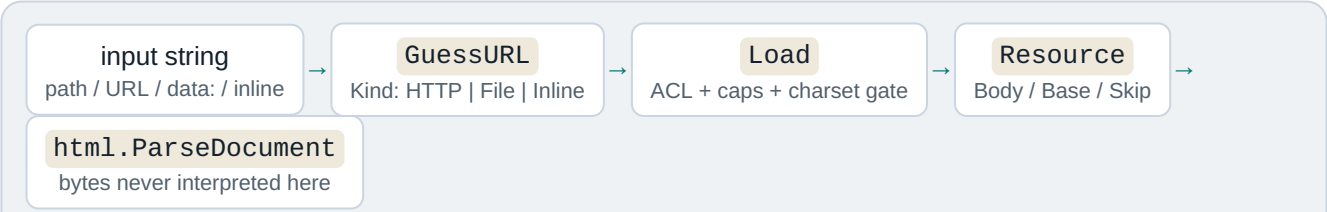
Known gaps: `default-encoding` is accepted-ignored; `size.pagesize` dual-write is transitional; object-level `web.background` is inert; the typed builder covers global PDF options only.

Position: upstream of every entry point — dotted Set/Get → UnitReal → page sizes → convert pageGeometry. Source: *documentation/architecture/03-settings.md* (commit ef526f9, 2026-08-11).

## 04 Load layer - URLs, I/O & ACL

Resource-fetching seam and trust boundary: URL guessing, local-file ACL, HTTP hardening, size caps.

`internal/load` owns the first pipeline stage and the two resource seams the engine depends on: the **primary load** (one `Loader.Load` per body object, cover, TOC, or header/footer) and the **subresource seam** ( `ResourceContext.Fetch` for linked CSS, images and `@font-face` URLs). It is the lowest internal package - only `internal/settings` and the Go standard library sit beneath it - and it carries the product security posture: default-deny local-file ACL, connect/response timeouts, a 10-redirect cap, a 100 MiB body cap, and a strict UTF-8/ASCII charset gate.




Primary load flow: an input string crosses the trust boundary into raw bytes.

### Primary load

`Loader.Load(ctx, input, pageLoad)` resolves pages, covers, TOCs and header/footer HTML like top-level documents. Budgets: 30 s connect, 60 s response default, 100 MiB cap, max 10 redirects, per-page `--timeout` override.

### Subresource seam

`ResourceContext.Fetch(ref)` binds a document base plus the per-page policy, so CSS, images and fonts inherit the same ACL and caps - e.g. `internal/convert/prepare/styles.go` collects linked stylesheets through this one narrow seam.

| | |
|---|---|
| `internal/load/load.go` | URL guessing, HTTP/file/data: fetching, cookie jar, proxy, auth, POST, ACL, caps, charset gate (1,322 lines) |
| `internal/load/load_test.go` | 30 behavioural tests: ACL matrix, body caps, timeouts, redirects, subresources, concurrency |
| `internal/load/doc.go` | Package doc: "reimplements the MultiPageLoader orchestration layer" |

- `Kind` - resolved input class: HTTP | File | Inline
- `Resource` - fetched document: URL, Base, Body, Skip
- `ResourceContext` - the one sanctioned subresource seam
- `AccessController` - default-deny ACL, symlink-resolved prefixes
- `NewLoaderWithError` - fail-fast constructor used at the request boundary

Local-file ACL decision matrix (THREAT-MODEL §3)

| Global enable | Object block | Read allowed |
|---|---|---|
| off | on (default) | no |
| off | off | no |
| on | on | no |
| on | off | yes |

URL shapes handled: `http://` `host` `/path`, `file://localhost/` `tmp` `/report.html` (remote file hosts refused), `data:` URIs, and bare input defaulting to `http://<input>` - a direct port of wkhtmltopdf's `guessUrlFromString`.

Trust boundary owner: every input string crosses here into raw bytes; nothing below `internal/load` touches the network or filesystem.

**Security:** default-deny ACL with `--allow` prefix expansion; no JavaScript ever; oversized bodies are rejected, never truncated; the TOCTOU window between ACL check and read is accepted residual risk; HTTP egress to localhost/RFC1918 is intentionally unrestricted (input-trust model). Full model: `documentation/THREAT-MODEL.md`.

First pipeline stage feeding `internal/html` + the `prepare` subresource seam · deep-dive: documentation/architecture/04-load.md

## 05 HTML parser: tokenizer & tree

Generic tokenizer and tree builder: entities, tolerant parsing, script content stripped at layout.

internal/html is the parse stage of the pipeline: a hand-rolled, *stdlib-only* leaf that owns the only DOM in the repo — every downstream stage walks its `Node` tree. Scanning is streaming: `scanTokens` emits tokens through a sink callback while the tree is built incrementally in the same pass, so a whole-token slice is never materialized.

Token constructs the scanner recognizes

`text run` `<!-- comment -->` `<!doctype>` `<?PI?>` (skipped) `</end>` `<start>` `<raw>` (verbatim)

Raw-text elements — `script`, `style`, `textarea`, `title` — are captured verbatim up to their real closing tag, then hidden by the UA sheet (`display:none`, `style_values.go:986`): script content is stripped at *layout*, not at parse. Entities decode once, at text-append and attribute application, via `UnescapeEntities` over stdlib `html.UnescapeString`.

A ruby annotation parses as one inline run — base, parens and reading are ordinary inline nodes (CJK ruby needs `--font-path` for glyphs):

XML(Extensible Markup Language) 2026-08-11 · html.go ≈ 832 lines · allowlistedbyconstruction

Tolerance is a feature: ~~browser-grade recovery~~ is deliberately out of scope, while malformed input degrades to a usable tree — six sentinel errors, never a panic.

| | |
|---|---|
| **internal/html/html.go** | Tokenizer, tree builder, Node model, walkers, element vocabularies (≈832 lines) |
| **internal/html/entities.go** | UnescapeEntities — &-gated stdlib entity decoding |
| **internal/html/html_test.go** | Same-package tests: tokenizer, tree, malformed-input table (≈901 lines) |
| **Consumers** | css.Match, layout style resolution, outline heading collection, convert HF/TOC/islands |

- `Parse` / `ParseDocument` — string vs bytes (+BOM) entry points
- `Node.Walk` — pre-order traversal used by every consumer
- `scanTokens` — streaming scanner with sink callback
- `autoClose` — implicit-close vocabulary (li, p, tr, td, th…)
- `mergeRootElement` — html/head/body merge instead of nesting
- `uaDecls` — UA sheet hiding script/style/title/head/meta/link

> Security: no script execution by construction — event-handler attributes are inert data; charset is enforced (UTF-8/ASCII only) at the load seam before parsing; hostile input cannot panic.

Position: load - **html** - (css matching, layout) · deep-dive: documentation/architecture/05-html-parser.md

## 06 CSS subsystem: parse, selectors & cascade

CSS subset: selectors, cascade, media and container queries; unknown rules degrade; :has and :target supported.

`internal/css` is a pure parsing, matching and value library that sits between the HTML tree and the layout engine. It owns the *CSS subset* the converter accepts and reports matches and specificity to `internal/layout`, which performs the cascade. It never fetches resources, never resolves page size, and never decides layout — the contract is quoted in its package doc (*css.go:1–18*).

- `parse-time` · `match-time` · `value helpers` · `never panics`

`html` **`css`** `layout` `pdf / image`

Supported surface: *`type`*, *`.class`*, *`#id`*, attribute selectors, `:first-child` / `:last-child` / `:nth-child`, `:has()`, `:not()`, descendant/child/sibling combinators, `@media`, `@container`, `!important`, and inline `style=""`. Everything else degrades without panicking — and unsupported selectors never silently degrade to the host element (`writePseudoLiteral` keeps unknown pseudos on the compound, so `li:target` can never paint every list item).

### Degrade rules — how the subset stays honest

- `garbage` is dropped: only unbalanced braces error, and `ParseNeverPanics` pins the no-panic property on adversarial input.
- `never-match` pseudos (`hover`, `active`, `focus`, `target`) stay on the compound and match `false` — a static PDF has no pointer state.
- `skip` at-rules: `@import`, `@supports`, `@layer`, `@keyframes`, nesting are silently skipped.
- `parse-early`: media preludes stay raw strings (the viewport is only known at conversion); `:nth-child` arguments compile to integer form so matching is pure arithmetic.

### Known gaps — what degrades today

**WIRE FORMAT — ONE DECLARATION BLOCK**

```
--accent: #b4532a;          custom property
color: var(--accent, #b4532a);   var() with fallback
font-size: 1.1rem !important;    single global tier
```

Unit math is tuned for IEEE float cancellation: `25.4mm` is computed as `val × 72 ÷ 25.4` so it cancels cleanly to `72pt` instead of `71.999…`; `px` maps at `0.75` (`96` `css px` `/in` → `72` `pt` `/in`); the media em base is `12` `pt`. Specificity is Selectors-4: `:has()` / `:not()` contribute the max of their arguments — a triple `(a, b, c)`[1] with *`a`* = ids, *`b`* = classes/attrs/pseudos, *`c`* = types.

[1] Functional pseudos contribute the specificity of their most specific argument, not a flat class-level count.

| | |
|---|---|
| **`internal/css/css.go`** | Sheet/rule/selector model, parser, at-rules, right-to-left matching, specificity (≈1784 lines) |
| **`internal/css/values.go`** | Inline style, `!important`, length/color/font parsing, custom-property resolution |
| **`internal/css/container.go`** | `@container` prelude, boolean condition tree, `LengthToPt` |
| **`internal/css/has.go` · `media.go`** | `:has()` / `:not()` combinator walk; `@media` evaluation |

- `css.Parse` — whole-sheet parse; only unbalanced braces error
- `css.Match / MatchPseudo` — selector vs node, right-to-left; `::before/::after` shapes

- `css.Specificity` — `(a,b,c)` triple; `:has()/:not()` max-of-arguments
- `css.MediaMatches` — raw `@media` prelude vs conversion viewport
- `css.ResolveCustomProps` — the single place custom-property policy lives
- `css.HasContainerRules` — fast gate; layout skips its second style pass when false
- `css.ParseSelectors` — strict list parse for `--exclude-from-outline`

> **Import rule:** `internal/css → internal/html → stdlib only`; css never imports layout, convert, pdf, load, or settings — viewport/media/container context is always passed as arguments.
> **Security:** css executes no code and fetches nothing; `url()` is honored only in `@font-face src` via the loader's ACL, and stylesheet rule budgets are capped in preparation (soft warn 25 000, hard cap 1 000 000).

Position: html - **css** - layout style resolution · measured 2026-08 · deep-dive: documentation/architecture/06-css.md

## 07 Layout engine & line breaking

Style cascade plus all formatting contexts, line breaking, pagination, emitting the display list.

internal/layout is the largest domain in the repo: it runs the cascade and builds an absolute-positioned *display list* of drawing ops on a continuous y-down canvas. Layout also owns pagination - page-break policies, thead repetition, sticky clamps, and the op-split machinery. One engine feeds three painters: PDF, header/footer bands, and the PNG/JPEG rasterizer.

`load` → `parse` → `css` → `layout` → `paginate` → `paint` → `write`

*layout and paginate both live in internal/layout; the same ops feed pdf.Paint, PaintBand (HTML header/footer) and the imageout rasterizer.*

### One display-list `Op` OP KINDS

Every op carries a stable `ID`. Kinds: `OpFillRect`, `OpStrokeRect`, `OpLine`, `OpText`, `OpImage`, `OpLinkURI`, `OpBullet`. Rects that split across a page boundary keep the source ID, so element-to-op ownership survives fragmentation.

Final z-order is resolved once by `PaintOrder` and shared by all three backends.

### Formatting contexts *(report-lite subset)*

- block & inline flow
- flex lite (row / column)
- multicol lite
- relative / absolute / fixed lite
- tables: colspan, rowspan, thead repeat
- grid lite + subgrid-inherit
- float lite + clear
- print sticky, 2D transforms

Key entry points (layout)

- `LayoutContext` / `WithWorkspace` - the layout passes
- `PaintContext` - paginate and paint into a `pdf.Document`
- `PaintBandContext` - single clipped band for HTML header/footer
- `CloneResult` - deep copy for the TOC page-count fixpoint

Scale: `~33.6k` lines incl. tests, `27` production files, `62` test files, `244` test functions.

| | |
|---|---|
| `internal/layout` | cascade, boxes, ops, pagination, paint; entry points LayoutContext / PaintContext |
| `internal/line` | naming note: log-line severity protocol (line.Emit), NOT text breaking - line breaking lives in layout inline.go / inline_collect.go |
| `internal/svg` | Rasterize is used for inline SVG images; layout stays one layer below convert and never imports convert/load/settings |

- `Result` - Ops, Pages [][]int, Locations []ElementLocation
- `ResolvedStyle` - ~1.3 KB per style, interned per run via styleStore
- `ElementLocation` - feeds outline.Lookup and #frag link assembly
- `Options` - Images func(src) callback keeps the ACL-gated loader injected
- `Workspace.Release` - display-list storage reuse for the page-islands path

Images arrive only through the injected Images callback (ACL-gated); layout never opens URLs and never runs JS. Work is bounded: the pagination fixpoint caps at 10 iterations, and SVG rasterizes at a 1024 px cap. Lite limits: positioning, sticky, float, flex/grid/multicol are report subsets; overflow is not general clipping; no vertical writing modes; HTML header/footer bands clip rather than paginate.

style - boxes - paginate - display list (consumed by pdf OR imageout) · deep-dive: *documentation/architecture/07-layout.md*

## 08 Convert pipeline — job orchestration

Orchestration hub: the Request union, 3-stage render.Pipeline, headers/footers, TOC fixpoint, outline, page islands.

internal/convert is the only package that coordinates a complete conversion. It owns the neutral `Request` union (convert.go:51), runs the per-object render loop, and drives every document-wide pass — TOC, outline, links, headers/footers, copies — before writing a single PDF. The lifecycle is delegated to a three-stage `render.Pipeline`, keeping PDF specifics out of the stage driver.


Lifecycle adapter: render.Pipeline (render/pipeline.go) owns stage order and cancellation; pdfPipeline (pdf_pipeline.go) owns the PDF detail. Cancellation is checked at every object boundary and between stages.

### Assemble, in order

1. `assembleTOC` — two-pass fixed point (`renderTOCObjects`, toc.go:221): `iteration 1` measures with tocGuess 0, `iteration 2` renumbers with the measured total and keeps the final layouts; pages reorder so the TOC comes first.
2. `assembleOutline` — pure heading tree (`internal/outline`, no PDF types) → `emitOutline` → `pdf.Outline` with final page refs.
3. `assembleLinks` — TOC forward/back links plus `#id` GoTo annotations (`applyInternalLinks`).
4. `assembleDocument` — page plan, title, compression, grayscale, creation time.
5. `assembleCopies` — collate / non-collate materialization via `render.Plan`.
6. `assembleHeadersFooters` — final pass; needs [topage] and section/subsection, which only exist once the whole document is final.

> Fact: headers/footers are painted once at the very end — after copies — because `[topage]` and section/subsection are only known when the whole document exists. Auto margins reserve the bands at layout time (`effectiveMargins`, hf.go:596); the draw pass clips ink into those same bands.

| | |
|---|---|
| **internal/convert** | job model, per-object loop, document-wide passes |
| **internal/convert/prepare** | shared load → parse → sheets → fonts phase (same seam used by image mode) |
| **internal/convert/render** | Pipeline lifecycle driver + Plan page-index model |
| **internal/outline** | pure heading tree, dump XML, section lookup (no layout/PDF result types) |

- `Request` — neutral pipeline input: Global, Image, Objects, Now, Output, OutlineOutput (convert.go:51)
- `render.Pipeline` — RenderObjects / Assemble / Finalize (render/pipeline.go:12)
- `runContext` — one conversion lifecycle: loader, fonts, single pdf.Document
- `objectState` — per-object geometry, HF, headings, navigation projection
- `pagePlan` — owner map + Remap after TOC reorder / copies (page_plan.go)

- `drawHeadersFootersResult` — final HF pass with real page numbers (hf.go:637)

**Security:** the loader (proxy/ACL boundary) is built at the top of `Run` before fonts or layout state; every subresource fetch inherits its ACL, timeouts and body caps. Header/footer values that look like HTML are treated as raw markup, never as URLs — blocking a local-path injection vector. Output sinks are caller-supplied writers; convert never opens paths. Budgets: 10 000 objects, 1 000 copies, 100 000 pages.

**Limits:** `--xsl-style-sheet` is not implemented (built-in TOC template, warning + fallback); the TOC fixed point is bounded to two iterations; HTML headers/footers are single-band clamps.

Progress is a (phase, percent) callback:  100 = progressComplete at Finalize; the CLI prints phase lines unless `--quiet`.

Position: Request → RenderObjects → Assemble → Finalize (drives every stage) · deep-dive: 08-convert-pipeline.md

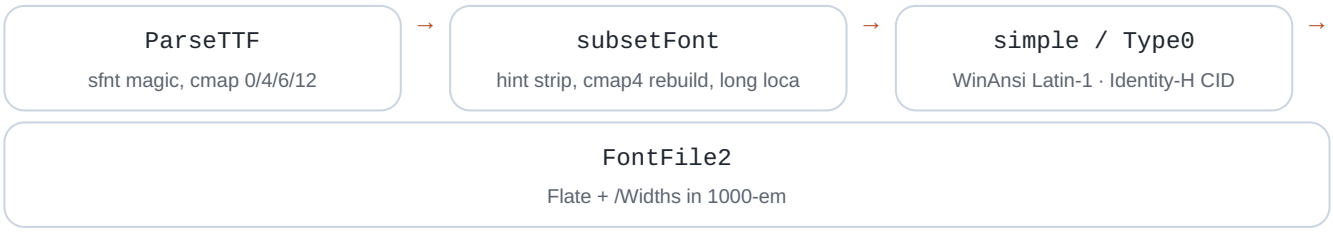
## 09 PDF 1.4 writer

PDF 1.4 sink: Flate streams, font subsetting, Type0/CID, xref, outlines, byte-stable output.

internal/pdf is the lowermost writer layer: it turns the layout display list into a viewer-openable PDF 1.4 file using only the Go standard library plus one allowlisted shaping module (go-text/typesetting). It owns the document object model and serialization, the content-stream operator language, the font pipeline (parse → subset → simple or Type0/CID embed), image embedding under hard caps, and outline/annotation wiring. Image mode reuses its fonts, shaping and glyph outlines but rasterizes instead of writing PDF bytes.

display list → paint → content streams → xref

Font pipeline: untrusted TTF / WOFF1 in, one subset per font per document out




### Content-stream language

Every operator is emitted by the `Content` builder: graphics state `q` / `Q`, paths `m` / `l` / `re` / `f` / `S`, transforms `cm`, and text `BT` / `ET` / `Tf` / `Tj`, with fonts and images registered into per-page /Resources at finalize. Text emits through TextShow: pure ASCII fast path, otherwise shape then split mixed Latin / CJK runs.

### Serialization

writeTo streams objects through a countingWriter so every xref offset is exact; all /FlateDecode streams use RFC 1950 zlib (raw DEFLATE made pages render empty). A typical tail:

```
0000000000 65535 f
0000000015 00000 n
0000000091 00000 n
trailer << /Size 3 /Root 1 0 R >>
```

Subsets are keyed by `fontCache[ v{0|1} | {fingerprint} | {baseName} | {runes} ]` so one deterministic subset per font per document is guaranteed, and hard caps bound untrusted input: `32 MiB` reconstructed WOFF, `16 384 px` image dimension, `16 Mpix` pixel budget.

| | |
|---|---|
| **internal/pdf** | Document model, operators, fonts, subsets, images, outlines (≈ 9.1k lines) |
| **internal/pdf/assets** | //go:embed Liberation + DejaVu faces; every accessor returns an isolated bytes.Clone |
| **internal/convert** | Caller: paints into Document, assembles TOC/outline/links/copies, then doc.Write |
| **internal/imageout** | Reuses fonts + shaping + glyph outlines; rasterizes instead of writing PDF bytes |

- `Document` - object model; AddPage, ReorderPages, DuplicatePage, SetOutline, Write
- `Content.TextShow` - ASCII fast path, else shape + split mixed runs; records runes for the subsetter
- `ParseTTF` - sfnt validation; rejects CFF/OTTO and WOFF2 (Brotli not allowlisted)
- `ensureFont` - dispatch simple vs Type0; one subset per font per document (cache key above)
- `emitType0` - Identity-H CIDFontType2; CIDToGIDMap kept uncompressed for viewer safety
- `finalize` - ordered wiring: outlines before catalog, pages tree, then xref + trailer

> **Security:** fonts are untrusted parse input — WOFF1 caps (tables ≤ 1024, per-table ≤ 16 MiB, overlap rejection), bounds-checked tables, no remote https @font-face fetch, system fonts opt-in only. **Determinism:** byte-identical output for identical input; CreationDate is injectable via Request.now() and otherwise pins a fixed date.

display list → paint → content streams → xref (right edge, PDF mode) · deep-dive: documentation/architecture/09-pdf-writer.md

## 10 Image output & SVG raster

Image-mode tail: rasterize the shared display list to PNG/JPEG; bitmap fonts; SVG via tdewolff/canvas.

Image mode shares the whole *load → parse → style → layout* front half with PDF mode, then diverges at the paint/write tail: `internal/imageout` paints the layout ***display list*** into an in-memory `image.NRGBA` canvas and encodes a single PNG or JPEG. `internal/svg` rasterizes `<img src="*.svg">` into PNG bytes through the allowlisted `tdewolff/canvas` path — the project's only third-party raster call.

`display list` → `rasterize 2× NRGBA` → `box-filter` → `encode PNG/JPEG`

Shared upstream, divergent sink — the Aug 2026 consolidation keeps text metrics, glyph forms and paint order on one table with PDF mode.

**PDF mode**

internal/pdf — Flate streams, font subsetting, xref

**Image mode**

internal/imageout — 2× supersample, no hinting, single page

| | |
|---|---|
| `internal/imageout` | ~3,000 lines: render options, supersampled rasterization, paint dispatch, decode/scale caches, PNG/JPEG encode |
| `internal/svg` | 450 lines: `svg.Rasterize` via tdewolff/canvas only; viewBox/width/height sizing, panic containment |
| `internal/convert/render` | Shared `RenderObjects → Assemble → Finalize` lifecycle; `Assemble` is a no-op for image mode |
| `internal/pdf` | Consumed, not owned: `ShapeRun` text shaping, `GlyphContours`, `DefaultFont`, `ScanFontDirs` |

- `RenderContext(ctx, root, opts)` — the core library render: layout result → rasterize → crop
- `RunRequest(ctx, req, log)` — engine seam shared by the `gowkhtmltoimage` CLI and `ImageConverter`
- `rasterizeContext` — paints `layout.Op` ops in `PaintOrder`, box-filters `rasterSS` supersample down
- `ttfDrawString` — TTF outline → bitmap glyphs (no FreeType, no hinting); fractional baseline, fake-bold pass
- `svg.Rasterize(data, maxSide)` — SVG → PNG bytes at intrinsic size; failure is a clean "no image"

Raster text uses ***TTF outline AA***— AA via 2× supersampling with a box-filter downscale; no hinting. Fidelity claims are pinned in *10-imageout-svg.md* and *fidelity.md*; the shape was consolidated 2026-08-11.

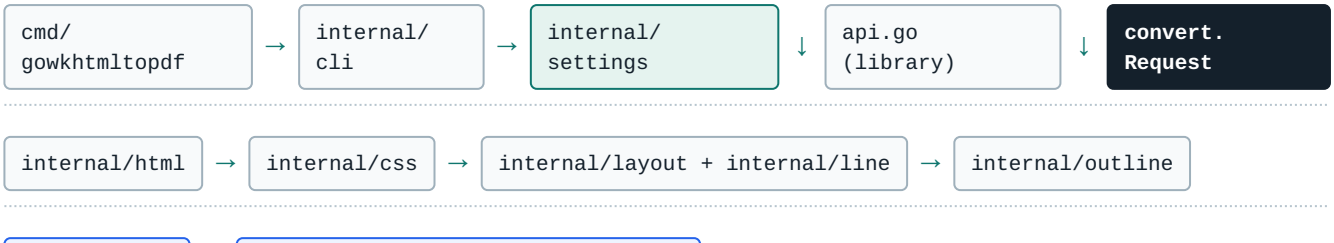
**Security & limits**

- ACL merged (`Image.Load` ⊕ `Global.Load`); subresource fetches capped (64 fetches / 32 MiB)
- Decode hardening: ≤32 MiB encoded, dims ≤16,384, ≤16M pixels; canvas ≤64M px / ≤256 MiB
- `--transparent` honored for PNG only; JPEG composites onto white with a warning
- Malformed SVG → clean error, never a crash (`defer recover()` in `rasterizeCanvas`)
- Single-page only: extra page objects and TOC are ignored with a warning
- Nearest-neighbour `<img>` scaling (Go 1.26 removed `image/draw` scalers)

Position: display list → raster → PNG/JPEG (right edge, image mode) · deep-dive: documentation/architecture/10-imageout-svg.md

## Dependency DAG a strict downward graph toward the trust boundary

Nothing points back up. `internal/cli` is a leaf; the engine never parses argv. Library, image binary and PDF binary funnel into one neutral seam: `convert.Request`. `internal/load` sits at the bottom as the trust boundary — every subresource crosses the same two seams and inherits the same ACL, timeout and size caps.




1. The engine never parses argv — the parser only writes dotted settings.
2. One neutral seam: `convert.Request` + `render.Pipeline` (RenderObjects → Assemble → Finalize).
3. `cli.Command` *is* the settings payload; layout/pdf never import settings.
4. Subresources cross `load.Loader.Load` / `load.ResourceContext.Fetch` only.
5. Image mode is not a parallel engine — same front half, divergent paint tail.

## PDF vs image mode same front half, different tail

### internal/pdf — PDF 1.4

- content streams + zlib `Flate`, xref/trailer
- embedded `Liberation Sans` subsets, /Widths 1000-unit em
- Type0/CID for CJK, WOFF handling
- Catalog outlines, link annotations, metadata
- byte-stable output when `Now` is injected

### internal/imageout — PNG/JPEG

- rasterize the shared display list — `Assemble` is a no-op
- bitmap font rasterizer (2× supersample, no hinting)
- single page; `--transparent` fill-alpha divergence
- SVG-as-image via `tdewolff/canvas` (only third-party raster call)
- reuses pdf shaping + outline data

## Security architecture default-deny, enforced at the seams

**ACL**
Local file access **denied by default**; opt-in via `--enable-local-file-access` / `--allow` prefixes, symlinks resolved.

**HTTP**
30 s connect / 60 s response timeouts, max 10 redirects, 100 MiB body cap on both sides.

**JS**
No JavaScript, ever — script content stripped at the layout stage.

**BUDGETS**
Rule and resource budgets enforced at the convert layer, not scattered in callers.

**HF**
Header/footer raw-markup rejected at the convert layer.

Full model: `documentation/THREAT-MODEL.md` · embedding guidance: `documentation/integration-security.md`.

Generated from `documentation/architecture/` — commit `ef526f9` (2026-08-11). Ten domain sections authored by ten parallel subagents; shell composed by the orchestrator. Pure HTML + CSS, no scripts.